Goal 2: A Frequent Relation Is Learned as Nonlinear

factual-recall-mechanisms

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1 Claim and result in one paragraph

A small transformer is trained for **exactly one epoch** on a synthetic corpus containing two relations with *identical frequency statistics at every level* (same subjects, same 32 candidate objects, same number of triples, same number of occurrences per triple): a semantically supported relation R_{sem} and a random relation R_{rnd}. At *high* fact frequency (F=64 occurrences per triple) both relations are *fully learned*: recall is 1.000 on all 128 training triples for both, in all three seeds. Yet an affine map fitted by ordinary least-squares linear regression from subject representations to object representations identifies held-out targets at 0.891 ± 0.056 for R_{sem} and at 0.016 ± 0.022 (= chance) for R_{rnd}. The frequent random relation is therefore stored, but stored *non-linearly*: no affine map over the same representations accounts for it, and it does not generalize (held-out recall at chance). Because the semantic relation at the very same frequency remains linear, the nonlinearity is attributable to the relation's semantics, not to its frequency; conversely, raising frequency from 1 to 64 occurrences never makes the random relation linear.

2 Data

Vocabulary (265 tokens). A separator `x`; two relation tokens `r_0`, `r_1`; 192 subject tokens `a_0`, ..., `a_191`; 32 object tokens `b_0`, ..., `b_31`; 32 attribute tokens `f_0`, ..., `f_31`; 6 attribute tokens `g_0`, ..., `g_5`.

Entities and attributes. Subject `a_i` carries the attribute pair $(f_{i \bmod 32},\ g_{\lfloor i/32 \rfloor})$, so the 192 subjects realize each of the 32×6 attribute combinations exactly once. Object `b_j` carries $(f_{(j+3) \bmod 32},\ g_0)$.

Sentences. All training sentences have five tokens:

attribute sentence	`x a_i f_p g_q x`	and	`x b_j f_p g_q x`
relation triple	`x a_i r_r b_t x`		

Relations.

$$R_{\text{sem}}(a_i) = b_{f(i)}, \qquad R_{\text{rnd}}(a_i) = b_{\rho(i)},$$

where $f(i) = i \bmod 32$ is the subject's own f-attribute index — so the semantic target is the object whose f-attribute is the subject's, shifted by 3 — and ρ is a seed-dependent balanced random assignment onto the same 32 objects, independent of the attributes. Concrete example (seed 19): `a_37` has attributes (`f_5`, `g_1`), so R_{sem}(`a_37`) = `b_5`, while R_{rnd}(`a_37`) = `b_27`; the training sentences are `x a_37 r_0 b_5 x` and `x a_37 r_1 b_27 x`.

Split and exact frequency matching. For every `f_`-value, 4 of its 6 subjects are *trained* and 2 are *held out* (128 trained, 64 held-out subjects; seed-dependent). ρ is constructed balanced within both groups. Consequently, under *both* relations: every trained subject occurs in exactly one triple per relation; every object is the target of exactly 4 trained and 2 held-out subjects; both relations contribute exactly 128 training triples, each repeated the same number of times (F). The two relations are indistinguishable by any frequency statistic; only the assignment semantics differ. Held-out subjects appear in attribute sentences but never in any relation triple. These invariants are asserted by `tests/test_linear_shift.py`.

3 Model and training

Model	causal transformer, 1 layer, hidden 64, 4 heads, context 8
Embeddings	input and output embeddings untied; all weights $\mathcal{N}(0, 0.02^2)$
	no attribute-based initialization; nothing frozen; no loss reweighting
Parameters	84,288
Corpus	every attribute sentence $\times A$=128 (= 28,672 sentences)
	+ 128 triples per relation, each $\times F$ ($F \in \{1, 4, 16, 64\}$)
Training	**exactly one epoch**: a single pass over the shuffled corpus
Steps	452 (F=1) to 704 (F=64) steps at batch size 64
Objective	unweighted next-token cross-entropy on all positions
Optimizer	AdamW, constant LR 10^{-3} after 5 linear warmup steps,
	weight decay 0, gradient clip 1.0
Seeds	19, 31, 47 (control data assignment, initialization, and data order)

Note that "frequent" here means frequent within a single epoch: at F=64 the corpus contains each triple 64 times, but the model still passes over the corpus exactly once.

4 The metric

A single analysis metric is used, as an operationalization of "the relation is an affine transformation":

> Fit $\hat{W}, \hat{b}$ by ordinary least-squares linear regression from the subject's *input embedding* $s_i \in \mathbb{R}^{64}$ to the target object's *output embedding* (unembedding row) $o_{R(i)} \in \mathbb{R}^{64}$, using the 128 trained pairs. For each of the 64 held-out subjects, predict $\hat{W}s_j + \hat{b}$ and select the nearest object row by cosine similarity among all 32 objects. Report the fraction of held-out subjects for which the selected object is $R(j)$. Chance is $1/32 \approx 0.03$.

For every subject exactly one object is correct under each relation, so no top-k metrics are needed or used. In addition, one *learning check* (not an analysis metric) is recorded: greedy next-token recall of the model on prompts `x a_i r_r`, for trained and held-out subjects. For the claim "learned as nonlinear" this check is essential: it establishes that the relation *is* stored.

5 Results

Finding 1: a fully learned frequent relation with no linear structure (F=64). At 64 occurrences per triple, R_{rnd} reaches perfect recall of all 128 training triples — the relation is fully stored — while the affine probe remains at chance (0.016 ± 0.022) and held-out recall is at chance (.031). The storage is a non-linear, non-generalizing memorization.

	probe accuracy (held-out)		recall (trained)		recall (held-out)	
F	R_{sem}	R_{rnd}	R_{sem}	R_{rnd}	R_{sem}	R_{rnd}
1	$1.000 \pm .000$	$.031 \pm .013$	.029	.010	.016	.010
4	$1.000 \pm .000$	$.016 \pm .013$	.055	.034	.057	.031
16	$0.990 \pm .007$	$.036 \pm .007$	.672	.167	.635	.031
64	$0.891 \pm .056$	$\mathbf{.016 \pm .022}$	**1.000**	**1.000**	.995	**.031**

Table 1: Mean $\pm$ sd over seeds 19/31/47, at A=128 attribute-sentence occurrences. F is the number of occurrences of each of the 128 triples per relation. Recall columns are the learning check. The headline cell is F=64: both relations are fully stored (recall 1.000), but only R_{sem} admits an affine map.

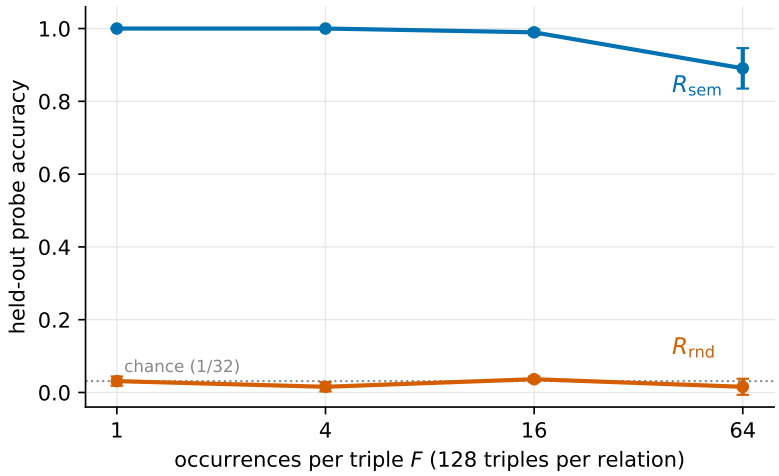


Figure 1: Held-out probe accuracy vs. occurrences per triple, for the two frequency-matched relations. The random relation is never linear, at any frequency; the semantic relation is linear at every frequency, with slight erosion at F=64.

Finding 2: the nonlinearity is due to semantics, not frequency. R_{sem}, trained in the same model from data with *identical* frequency statistics, is equally fully stored (recall 1.000) but linear (probe 0.891 ± 0.056) and generalizing (held-out recall .995). Conversely, increasing the frequency of R_{rnd} from 1 to 64 occurrences never produces linear structure (Fig. 1): frequency determines only *whether* the random relation gets stored (recall .010 $\rightarrow$ 1.000), never *in what format*.

Finding 3: high frequency slightly erodes the linear format of the semantic relation. R_{sem}'s probe declines from 1.000 ($F \leq 4$) to $0.891 \pm .056$ (F=64), consistent with high-frequency memorization beginning to compete with the linear solution, while remaining far from the random relation's chance level.

6 Learned representation and exact algorithm

The random facts are stored as a subject-indexed memory, not as a shared relation rule. Every subject token has its own learned embedding. Repeated random triples write subject-specific target

information into that embedding, while shared key/value projections and the output readout learn how to retrieve it. The embedding is a multiplexed record: it still carries the attribute-supported semantic code, but it also contains an idiosyncratic payload for $\rho(i)$. Because the subject's value vectors are computed before the following relation token is read, the values themselves are relation-independent; `r_0` and `r_1` select different uses of the same subject record through their queries and downstream residuals.

This representation is visibly different from Goal 1. A target centroid fitted on one target-stratified half of the memorized subjects decodes $\rho(i)$ from the other half's raw subject embeddings with only $.141 \pm .058$ accuracy (chance .031), compared with $1.000 \pm .000$ for the shared semantic code in Goal 1. The random target becomes increasingly explicit as the record is read: centroid accuracy is $.766 \pm .175$ over the concatenated head outputs and $1.000 \pm .000$ after their projection. No corresponding payload is learned for subjects absent from relation triples, so both held-out recall and the held-out affine probe remain at chance.

The computation on a trained subject is equivalent to the following lookup procedure:

```
def answer_random(subject):
    # Learned representation: one memory record per trained subject.
    record = embedding[subject]              # key + target payload

    # r1 selects the random-relation read. H1 is primary;
    # H0/H2 provide seed-dependent support and H3 is negligible.
    retrieved = sum(head[h].read_value(record, relation=r1)
                    for h in [H0, H1, H2])

    state = relation_state[r1] + project(retrieved)
    state += distributed_mlp_correction(state)
    object_scores = dot(final_normalize(state), object_directions)
    return argmax(object_scores)              # b[rho(subject)]
```

Each `read_value` multiplies a projection of the subject record by that head's attention weight, and the final `dot` compares the retrieved state with the 32 learned object-output directions. This is a genuine causal lookup. Replacing all head outputs by their mean destroys the subject identity and reduces accuracy from $1.000 \pm .000$ to chance. Swapping the head outputs between two subjects transfers the clean subject's target-logit effect completely. Swapping only the subject-derived value payload transfers $.975 \pm .020$ of that effect; swapping only the two subjects' attention weights transfers $.023 \pm .014$. The key operation is therefore "read the content stored in this subject's value projections," not "route each subject along a unique attention pattern."

The relation token is the memory-bank selector. With the subject fixed, replacing `r_0` by `r_1` changes both the retrieval and its interpretation: head-output patching restores $.922 \pm .024$ of the relation effect and MLP-population patching restores $.841 \pm .032$ (the effects overlap and are not additive). H1 is the stable primary retrieval head, H0/H2 supply seed-dependent support, and H3 has approximately zero causal contribution.

Most lookup is complete before the MLP: the object is already correct at $.927 \pm .052$ after attention and at $1.000 \pm .000$ after the MLP. The latter is a distributed correction rather than a bank of individual fact neurons: keeping only the 32 highest-ranked variable units recovers $.478 \pm .022$ of its subject-specific margin, while ablating them leaves $.826 \pm .014$. An affine surrogate for all GELUs still obtains $1.000 \pm .000$ on disjoint memorized facts. Thus "nonlinear" describes the storage format—there is no affine rule that transfers ρ to new subjects—not a requirement for GELU curvature during each trained-subject lookup.

The frequent random relation is therefore recalled by a non-generalizing key–value memory: relation-gated heads retrieve a subject-specific value code, and a distributed MLP/readout sharpens it into the memorized object direction. There is no shared rule for subjects outside the memorized set, explaining the simultaneous perfect training recall, chance held-out recall, and chance held-out affine probe.

7 What this does and does not show

- **Shown:** a frequent relation can be fully learned and stored non-linearly (no affine map over the representations accounts for it, and it does not generalize); with frequency exactly matched, storage format is decided by semantics.
- **Qualification:** "frequent ⇒ nonlinear" holds for the relation without attribute support. Frequency alone does not force nonlinearity: the frequent *semantic* relation remains largely linear (with mild erosion at F=64). The precise claim supported is: at matched high frequency, an attribute-supported relation is stored linearly and an unsupported one non-linearly.
- **Superseded:** earlier drafts' high-frequency results used attribute-initialized embeddings (`structured_embedding_init`), targeted-row updates, and top-k metrics; they are replaced by the present end-to-end setup with a single OLS metric.

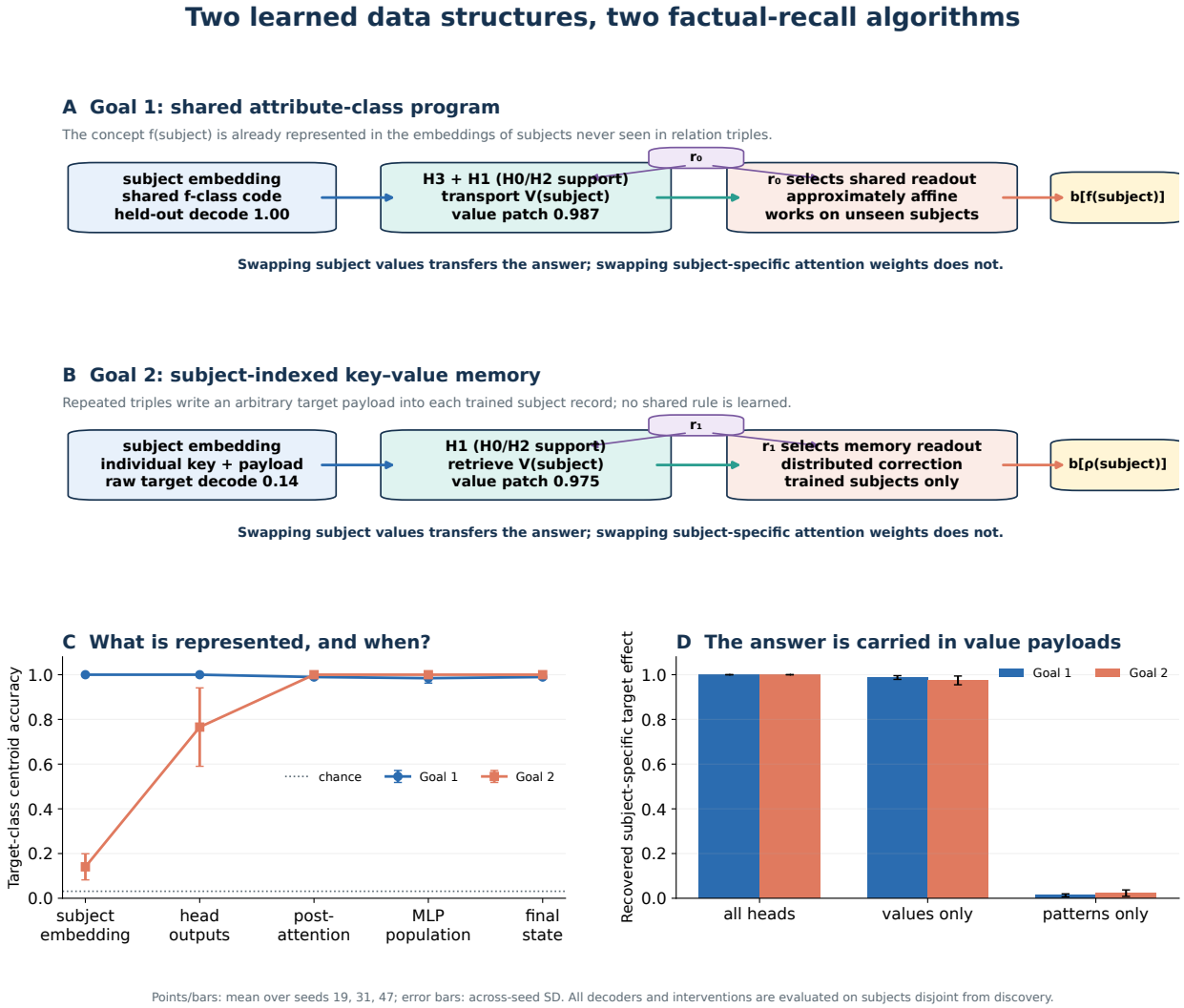


Figure 2: The learned data structures and equivalent algorithms for both goals. Goal 1 transports a shared f-class code into an approximately affine class-to-object readout; Goal 2 retrieves a subject-specific memory payload. The lower panels show how target identity becomes decodable and establish that subject information is carried by attention values rather than subject-specific attention weights.